9C Lesson 3

Control systems: nerves and hormones

Year 9 Science · 9C Threat from disease · Student Book pages 42 to 43

Name _____ Class _____ Date _____

Today I can:

- ☐ I can describe how the nervous system detects a stimulus and produces a response.
- ☐ I can describe how hormones affect the body and compare the nervous and hormonal systems.

Key words

nervous system the organ system that senses changes and makes the body respond rapidly

central nervous system the brain and spinal cord (CNS)

nerve an organ that carries impulses between the CNS and the rest of the body

stimulus a change that can be detected, e.g. light, sound, touch

receptor a cell in a sense organ that detects a stimulus

impulse an electrical signal that travels along a nerve

effector a muscle or gland that carries out a response

hormone a chemical messenger made in a gland and carried in the blood

gland an organ that makes and releases a substance, e.g. a hormone

target organ an organ that responds to a particular hormone

insulin a hormone from the pancreas that makes cells take up glucose from the blood

● Core Everyone does this section.

1. Match each sense organ to the stimulus it detects. Draw a line.

eye	chemicals in the air
ear	light
skin	chemicals in food
tongue	sound
nose	temperature and touch

2. Complete the text using words from the word bank.

The chemical messengers in our body are called _____. They are made in _____ and released into the _____ to travel round the body. They cause _____ organs to change what they are doing.

3. Ruler drop: record the distance the ruler fell before you caught it. Then calculate the mean.

a) Complete the table.

Try	Distance the ruler fell (cm)
1	
2	
3	
4	
5	
mean	

b) Which control system did you use to catch the ruler? Give a reason.

4. Number the statements 1 to 6 to show the path that impulses take when you pick up a pen.

Statement	Order
the brain processes the information	
touch receptors in the skin of the fingers detect the pen	
impulses travel along a nerve to the spinal cord and brain	
the muscles contract and lift the hand	
impulses travel along a nerve to the muscles in the arm	
the pen touches the fingers (the stimulus)	

●● Stretch When Core is finished.

S1. Name the organ that makes insulin and describe the effect insulin has on muscle cells.

S2. Complete the table to compare the two control systems.

	Nervous system	Hormonal system
type of messenger		
speed of response		
acts on		

S3. When we see something dangerous, adrenaline is released into the blood. Explain how this response prepares the body for dealing with danger.

●●● Challenge Explain in full sentences.

C1. You tread on something sharp. Suggest which control system is better for responding to this stimulus. Explain your answer, naming the receptor, the parts of the nervous system and the effector involved.

C2. Sultan Kosen grew to 251 cm tall. Chandra Dangi was 55 cm tall as an adult. Suggest which two glands may not have been working normally in these men, and explain your reasoning.

My thinking (any language)

The chickenpox virus can hide inside nerve cells for many years and later cause shingles. Why might white blood cells find it hard to destroy a virus hiding inside a cell?

Word bank stimulus · receptor · impulse · nerve · spinal cord · brain · effector · hormones · glands · blood · target

✂ ---------- cut here

Exit ticket 9C Lesson 3: Control systems: nerves and hormones Name _____

1. Name the two organs of the central nervous system.

2. What is a hormone, and how does it travel round the body?

3. Give one difference between a nervous response and a hormonal response.

Today I can: Goal 1 ☐ Goal 2 ☐